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4.3 SSS and SAS Congruence

Two triangles are congruent when there is a correspondence between their vertices such that all corresponding sides and angles match. Rather than checking all six parts, shortcut postulates let us prove congruence from partial information. The **SSS Postulate** says three pairs of congruent corresponding sides are enough. The **SAS Postulate** says two pairs of congruent sides plus the *included* angle (the angle directly between those two sides) are enough. The angle's position matters: if the given angle is not between the two given sides (the "SSA" case), the triangle is not guaranteed to be unique — this is why SSA is *not* a valid congruence shortcut. Shared sides use the Reflexive Property, and intersecting segments often supply a hidden congruent angle pair via the Vertical Angles Theorem. This worksheet applies SSS and SAS with numeric, algebraic, and coordinate reasoning.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. What does SSS stand for?

2. What does SAS stand for?

3. In SAS, where must the given angle be located?

4. Three pairs of corresponding sides are congruent. What can you conclude, and by what postulate?

5. Two pairs of sides and the included angle are congruent. Conclusion and postulate?

6. Two sides and a non-included angle are congruent between two triangles. Can you conclude congruence by SAS?

7. Two triangles share a side. What property justifies that the side is congruent to itself?

8. $\triangle ABC \cong \triangle DEF$. Side $\overline{AB}$ corresponds to which side?

9. $\triangle ABC \cong \triangle DEF$. $\angle C$ corresponds to which angle?

10. All three pairs of sides are marked congruent. Which postulate applies?

11. Two pairs of sides and the included angle are marked congruent. Which postulate applies?

12. In a congruence statement like $\triangle ABC \cong \triangle DEF$, does the order of the letters matter?

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13. $\triangle ABC$ has vertices $A(0,0)$, $B(6,0)$, $C(0,8)$. $\triangle DEF$ has vertices $D(0,0)$, $E(0,-6)$, $F(8,0)$. Compute all three side lengths of each triangle and determine whether they are congruent.

14. Segments $\overline{AC}$ and $\overline{BD}$ intersect at E. $AE = 3x - 1$, $CE = 11$, $BE = 5x - 6$, $DE = 14$. Find x, then explain why $\angle AEB \cong \angle CED$, and name the postulate proving $\triangle AEB \cong \triangle CED$.

15. $\triangle ABC \cong \triangle DEF$ by SAS, with $\angle A = 48°$, $\angle B = (2x + 10)°$, and $\angle E = 76°$. Find x, then find $\angle C$ and $\angle F$.

✓ Answer Key

1. Side-Side-Side
2. Side-Angle-Side
3. Between the two given sides (included)
4. The triangles are congruent, by SSS
5. The triangles are congruent, by SAS
6. No — the angle is not included, so SAS does not apply (SSA is not a valid shortcut)
7. Reflexive Property
8. $\overline{DE}$
9. $\angle F$
10. SSS
11. SAS
12. Yes — it identifies the correspondence between vertices
13. $AB = DE = 6$, $BC = EF = 10$, $CA = FD = 8$ — congruent by SSS despite the different orientation
14. $x = 4$; $AE = CE = 11$, $BE = DE = 14$; $\angle AEB \cong \angle CED$ by the Vertical Angles Theorem; congruent by SAS
15. $x = 33$; $\angle B = \angle E = 76°$; $\angle C = \angle F = 56°$